

A new LAPPD geometry model for WCSim

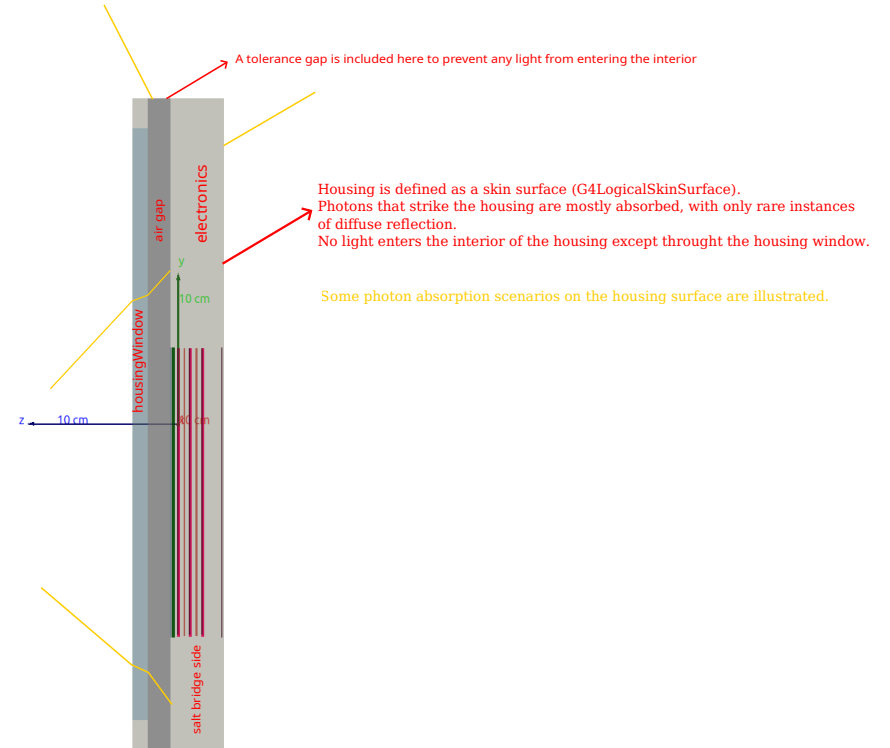
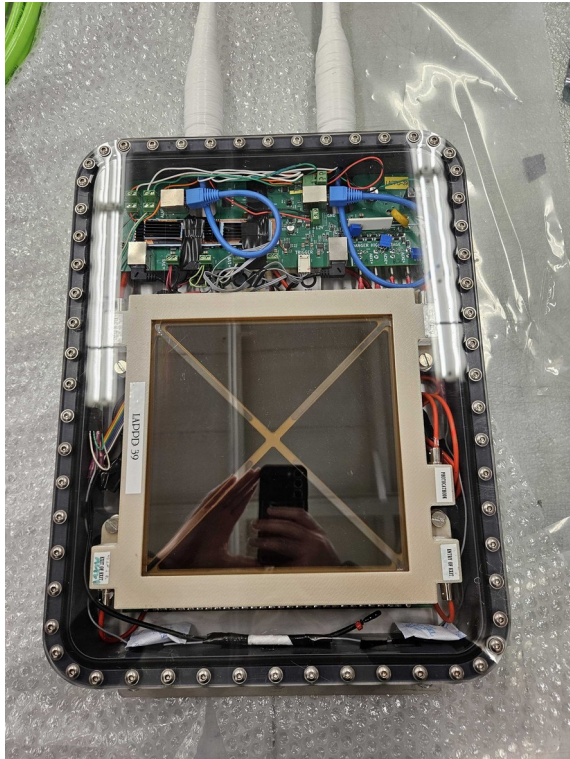
A Standalone LAPPD Simulation Package Using Geant4

- Designed to simulate LAPPDs in various detector environments
- Modular and application-independent
- Easy integration into any Geant4-based simulation
- Only requires defining and assigning a sensitive detector
- Source code: github.com/mkandemirr/LAPPD

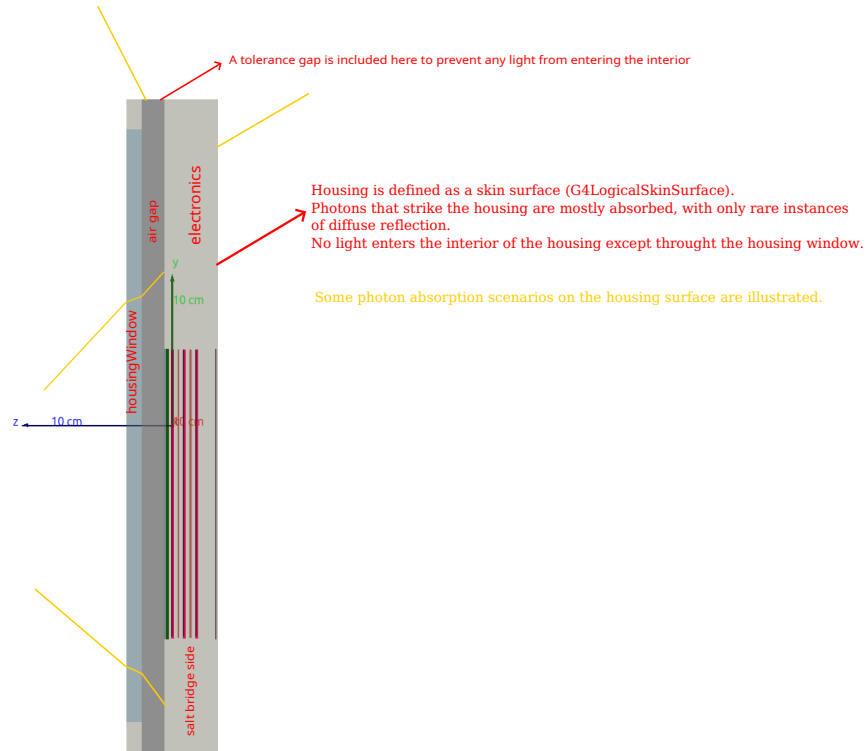
Used in Three Different Geant4 Applications

- A Toy Rectangular Prism-Shaped Cherenkov Detector
 - Included in the package to demonstrate how to use the LAPPDs in a simple Geant4 application
- Custom ANNIE Application
 - Uses the LAPPD package in a realistic detector context
- WCSim Integration
 - Only essential parts of the package were integrated
 - Adapted to fit WCSim's internal design and architecture

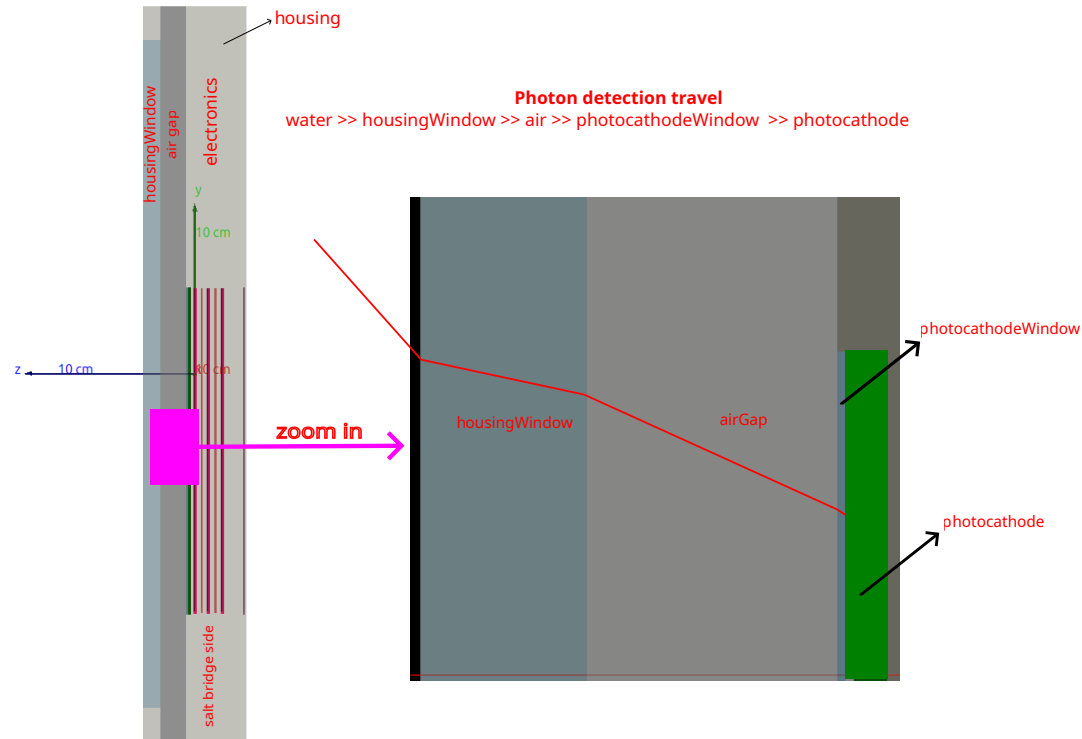
LAPPD – Real Device vs Simulation model



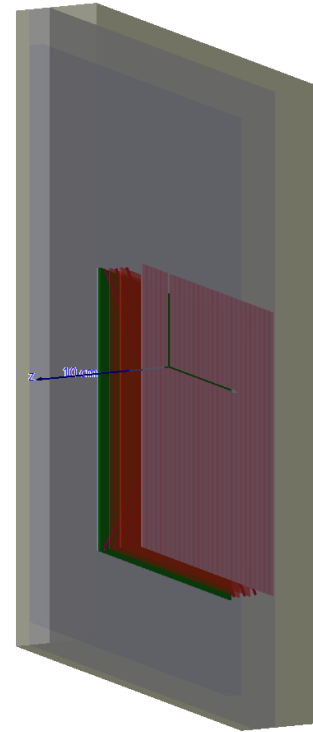
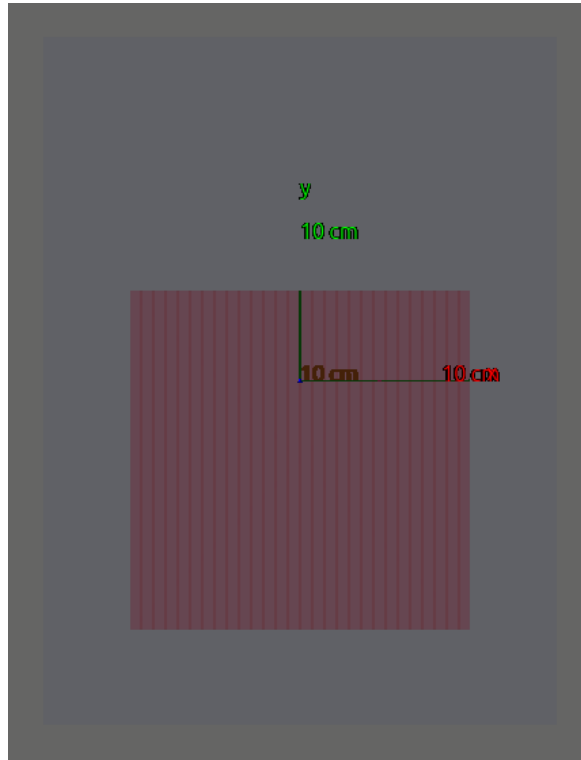
LAPPD housing skin surface



Photon detection travel

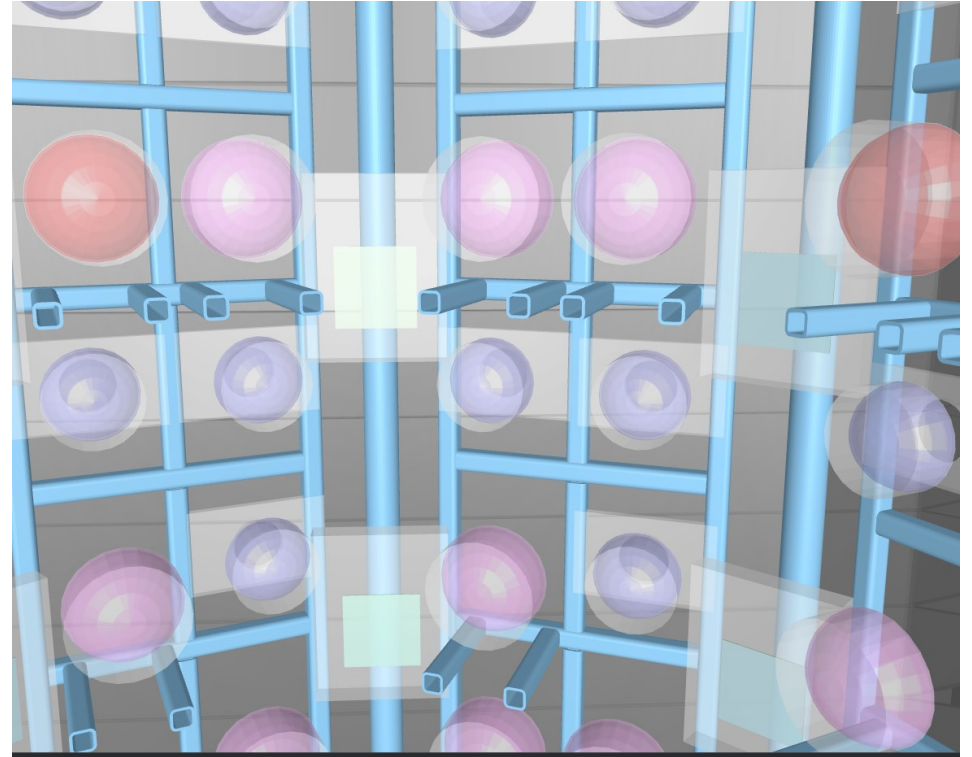
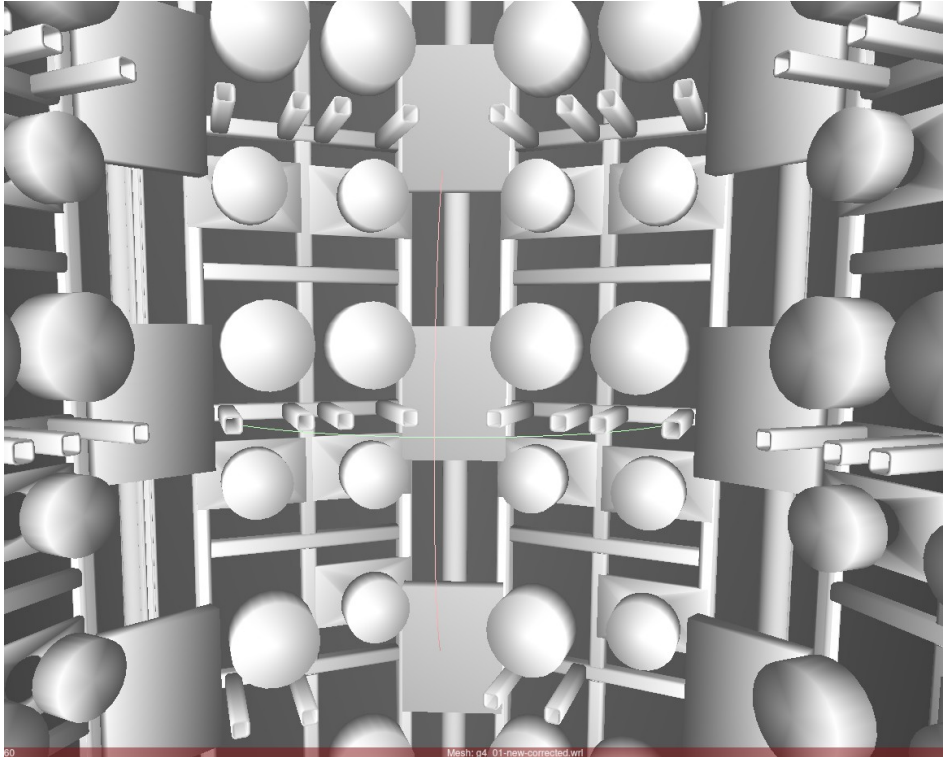


LAPPD geometry

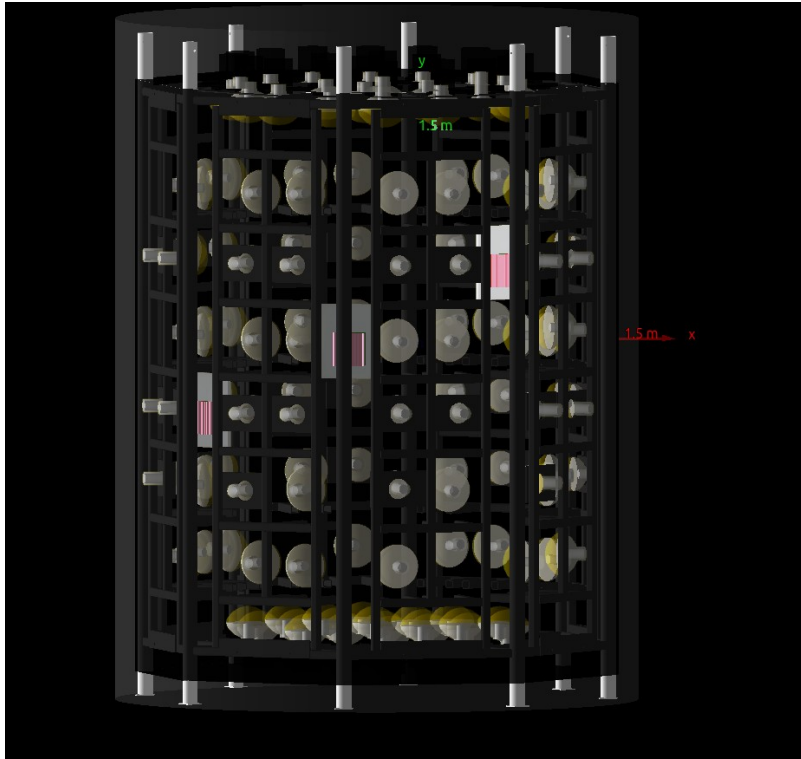


LAPPDs in 3 different g4-application (1)

WCSim

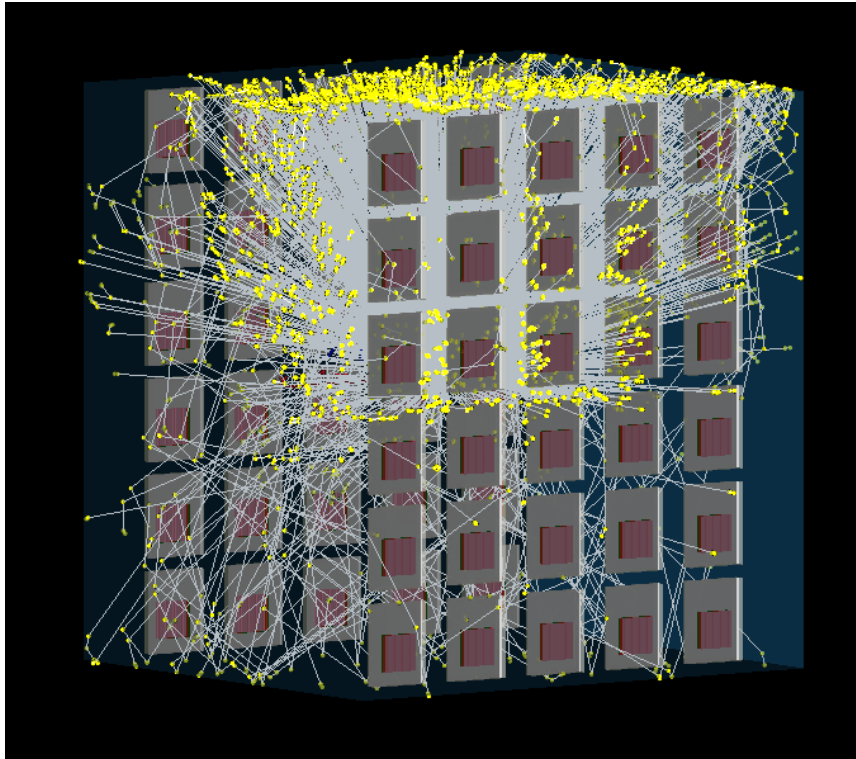


LAPPDs in 3 different g4-application (2)



- This anne-g4-application developed by myself. LAPPD package is easily integrated into it.

LAPPDs in 3 different g4-application (3)



- This application is included in the package to demonstrate how LAPPDs are used.

User interface

```
#lappdGeoSettings
/lappd/includeStrips true #just for visualization

/lappd/housingWindowMatName glass
/lappd/photocathodeWindowMatName borosilicateGlass

/lappd/photocathodeSizeX 191.5 mm
/lappd/photocathodeSizeY 191.5 mm

/lappd/housingWindowSizeX 290 mm
/lappd/housingWindowSizeY 390 mm
/lappd/housingWindowThickness 10 mm # z direction

/lappd/housingSizeX 330 mm
/lappd/housingSizeY 430 mm
/lappd/housingThickness 60 mm # z direction #when strips included, if you see overlaps just increase this value!

/lappd/airGapThickness 15 mm # z direction
/lappd/photocathodeWindowThickness 0.5 mm # z direction

#photocathode position with respect to its mother volume, namely, housing
/lappd/photocathodePositionX 0 mm
/lappd/photocathodePositionY -45 mm
```

WCSim: A photon detection history (1)

```
Photon at Boundary!  
thePrePV: WCLAPPD  
thePostPV: lappd_housingWindowPV  
Old Momentum Direction: (-0.0177166,0.136365,0.9905)  
Old Polarization: (-0.606545,0.786082,-0.119071)  
New Momentum Direction: (-0.168143,0.113692,0.979184)  
New Polarization: (-0.603481,0.773557,-0.193445)  
*** FresnelRefraction ***  
G4VDiscreteProcess::PostStepGetPhysicalInteractionLength [ OpBoundary]  
Particle type - opticalphoton  
  mass:      0[GeV]  
  charge:    0[e]  
  Direction x: -0.168143, y: 0.113692, z: 0.979184  
  Total Momentum = 2.10628e-09[GeV]  
  Momentum: -3.54154e-10[GeV], y: 2.39467e-10[GeV], z: 2.06243e-09[GeV]  
  Total Energy = 2.10628e-09[GeV]  
  Kinetic Energy = 2.10628e-09[GeV]  
  MagneticMoment [MeV/T]: 0  
  ProperTime = 0[ns]  
  in Material Glass  
InteractionLength= 1.79769e+307[cm]
```

WCSim: A photon detection history (2)

```
Photon at Boundary!  
thePrePV: lappd_housingWindowPV  
thePostPV: lappd_airGapPV  
Old Momentum Direction: (-0.168143,0.113692,0.979184)  
Old Polarization: (-0.603481,0.773557,-0.193445)  
New Momentum Direction: (0.398199,0.181855,0.899092)  
New Polarization: (-0.625828,0.770466,0.121335)  
*** FresnelRefraction ***  
G4VDiscreteProcess::PostStepGetPhysicalInteractionLength [ OpBoundary]  
Particle type - opticalphoton  
mass: 0[GeV]  
charge: 0[e]  
Direction x: 0.398199, y: 0.181855, z: 0.899092  
Total Momentum = 2.10628e-09[GeV]  
Momentum: 8.38716e-10[GeV], y: 3.83036e-10[GeV], z: 1.89374e-09[GeV]  
Total Energy = 2.10628e-09[GeV]  
Kinetic Energy = 2.10628e-09[GeV]  
MagneticMoment [MeV/T]: 0  
ProperTime = 0[ns]  
in Material AirGap  
InteractionLength= 1.79769e+307[cm]
```

WCSim: A photon detection history (3)

```
Photon at Boundary!  
thePrePV:  lappd_airGapPV  
thePostPV: ANNIep2v7-glassFaceWONLYLAPADS → Sensitive volume  
Old Momentum Direction: (0.398199,0.181855,0.899092)  
Old Polarization:      (-0.625828,0.770466,0.121335)  
New Momentum Direction: (-0.168143,0.113692,0.979184)  
New Polarization:      (-0.725612,0.65809,-0.20101)  
*** FresnelRefraction ***  
G4VDiscreteProcess::PostStepGetPhysicalInteractionLength [ OpBoundary]  
Particle type - opticalphoton  
  mass:      0[GeV]  
  charge:    0[e]  
  Direction x: -0.168143, y: 0.113692, z: 0.979184  
  Total Momentum = 2.10628e-09[GeV]  
  Momentum: -3.54154e-10[GeV], y: 2.39467e-10[GeV], z: 2.06243e-09[GeV]  
  Total Energy   = 2.10628e-09[GeV]  
  Kinetic Energy = 2.10628e-09[GeV]  
  MagneticMoment [MeV/T]: 0  
  ProperTime     = 0[ns]  
  in Material    Glass  
InteractionLength= 1.79769e+307[cm]
```

WCSim: A photon detection history (4)

```
Photon at Boundary!  
thePrePV: ANNIEp2v7-glassFaceWONLYLAPPDS  
thePostPV: lappd_photocathodePV  
Old Momentum Direction: (-0.168143,0.113692,0.979184)  
Old Polarization: (-0.725612,0.65809,-0.20101)  
New Momentum Direction: (-0.168143,0.113692,0.979184)  
New Polarization: (-0.725612,0.65809,-0.20101)  
*** Absorption ***  
G4VDiscreteProcess::PostStepGetPhysicalInteractionLength [ OpBoundary]  
Particle type - opticalphoton  
mass: 0[GeV]  
charge: 0[e]  
Direction x: -0.407937, y: -0.23974, z: 0.880972  
Total Momentum = 3.38582e-09[GeV]  
Momentum: -1.3812e-09[GeV], y: -8.11718e-10[GeV], z: 2.98281e-09[GeV]  
Total Energy = 3.38582e-09[GeV]  
Kinetic Energy = 3.38582e-09[GeV]  
MagneticMoment [MeV/T]: 0  
ProperTime = 0[ns]  
in Material Doped Water  
InteractionLength= 1.79769e+307[cm]
```

Recorded as hit

What is next?

- WCSim is now ready to run with the new LAPPD geo/optical model.
- A comparison study would be useful to observe the differences.

Written and updated files in WCSim

- A new class *WCSimConstructANNIELAPPD.cc* is written.
- *WCSimDetectorConstruction.hh* (added a few lines)
- *WCSimDetectorConfigs.cc* (added a few lines)
 - user only use this file to select one of the LAPPDGeo model
- *WCSimConstructANNIECylinderScan.cc*
- When the LAPPD housing with its actual dimensions was placed in the existing position, it overlapped with the ANNIE holders. With this correction, the LAPPDs have been shifted slightly forward, and now there is no overlap.
 - To prevent overlap, an optimized position correction ratio of 0.93 was applied.
- If you search “*added by MK*” in the updates files, you can easily find the location of the added lines.

How to use the new LAPPD model?

- 1 Copy the WCSim source from the following location to your own workspace:
 - `/exp/annie/app/users/kandemir/WCSim`
- 2 Compile the code.
- 3 Open the file *WCSimDetectorConfigs.cc*.
- 4 Inside the *WCSimDetectorConstruction::SetANNIEPhase2Geometryv7()* function, select the desired LAPPD geometry model by setting the following line:
 - `SelectedLAPPDModel = LAPPDGeoModel::ANNIE; // the new model`
 - `//SelectedLAPPDModel = LAPPDGeoModel::Default; // the existing model`

A comparison of the existing and new LAPPD geo model

